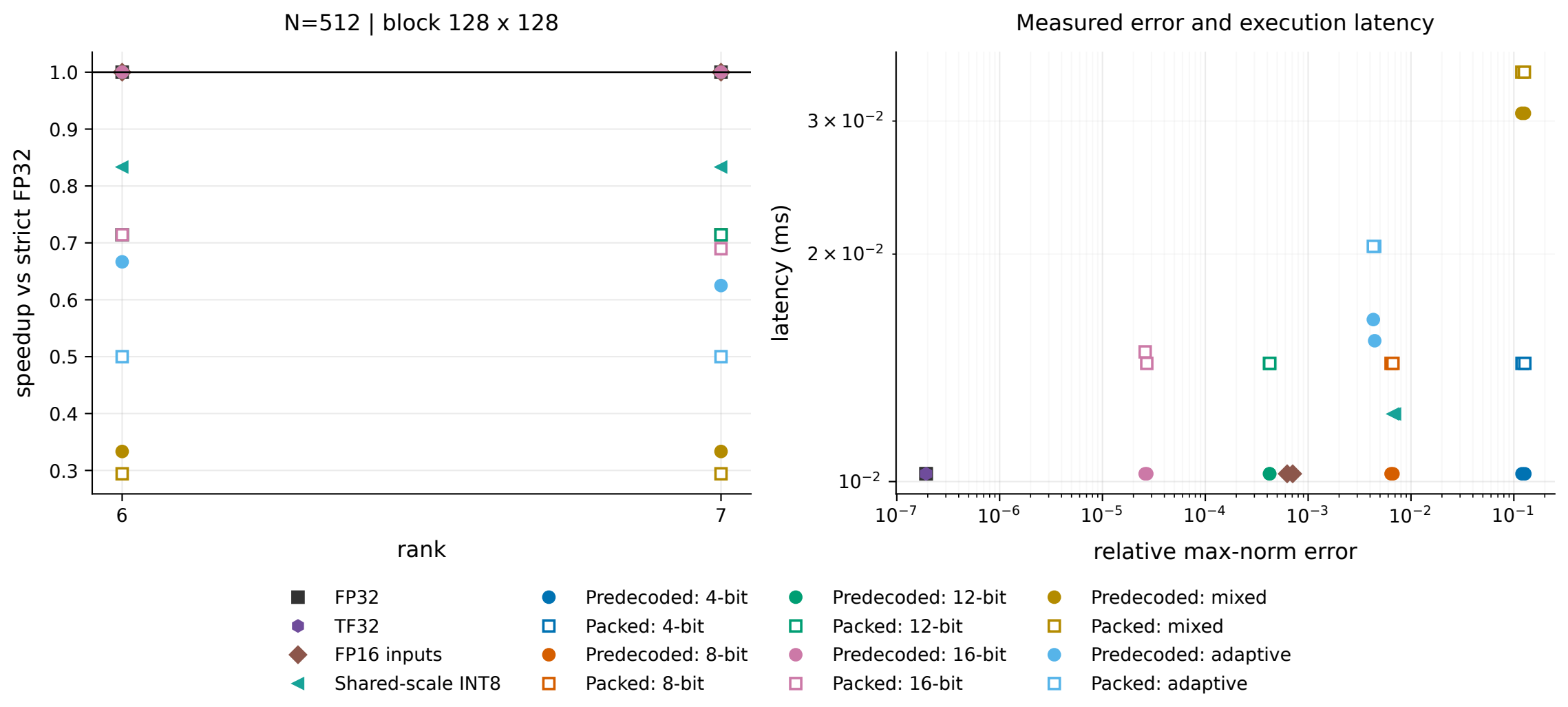
N=512 | block 128 x 128
speedup vs strict FP32
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
6
7
rank
Measured error and execution latency
latency (ms)
3×10^{-2}
2×10^{-2}
10^{-2}
10^{-7}
10^{-6}
10^{-5}
10^{-4}
10^{-3}
10^{-2}
10^{-1}
relative max-norm error
FP32
TF32
FP16 inputs
Shared-scale INT8
Predecoded: 4-bit
Packed: 4-bit
Predecoded: 8-bit
Packed: 8-bit
Predecoded: 12-bit
Packed: 12-bit
Predecoded: 16-bit
Packed: 16-bit
Predecoded: mixed
Packed: mixed
Predecoded: adaptive
Packed: adaptive